1 Vector space → 2 Subspace → 3 Linear combination →
4 Generating system → 5 Dependence → 6 Rank → 7 Basis and dimension

Each step uses the one before it. Then: what you need to know, check questions with answers.

STEP 1 · THE PLAYGROUND

Vector space

The setting for everything else on this page.

Intuition. A vector space is any set where you can add two elements and stretch one by a real number, and both operations behave the way arrows in the plane do. Its elements are called vectors, even when they are polynomials or functions.

DEFINITION

A set V with a sum $+$ and a product by real numbers $\cdot$ is a real vector space $(V, +, \cdot)$ if, for all $u, v, w \in V$ and $k, h \in \mathbb{R}$:

Sum	Product by a scalar
$(u+v)+w = u+(v+w)$	$k\cdot(u+v) = k\cdot u + k\cdot v$
$u+v = v+u$	$(k+h)\cdot u = k\cdot u + h\cdot u$
there is a 0 with $u+0=u$	$k\cdot(h\cdot u) = (k\cdot h)\cdot u$
each u has an opposite $-u$ with $u+(-u)=0$	$1\cdot u = u$

Examples. $\mathbb{R}^n$ with the usual operations; $\mathbb{R}_n[x]$, the polynomials of degree at most n;

the continuous functions on an interval I.

Always true (proposition). $0 \cdot u = 0$, $(-1) \cdot u = -u$, $k \cdot 0 = 0$, and $k \cdot u = 0 \Leftrightarrow k = 0$ or $u = 0$.

IN CODE

```
interface VectorSpace<T> {
  zero: T;
  add(a: T, b: T): T;
  scale(k: number, a: T): T;
}
// plus the 8 laws above, which no type checker can enforce
```

STEP 2 · USES STEP 1

Vector subspace

A smaller vector space living inside a bigger one.

Intuition. $\mathbb{R}^3$ is a world where you can add vectors and stretch them. A subspace is a smaller world inside it that is closed: whatever you add or scale, you never fall out of it. In TypeScript terms, a type where `add(a, b)` and `scale(k, a)` always return that same type.

DEFINITION

Let $(V, +, \cdot)$ be a real vector space. W is a vector subspace of V, written $W \leq V$, if:

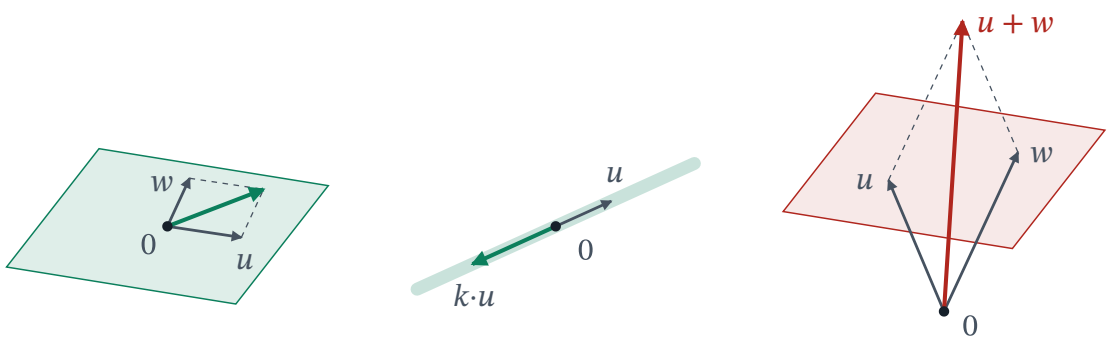
1. $\varnothing \neq W \subseteq V$
2. for all $u, w \in W$: $u + w \in W$
3. for all $u \in W$ and $k \in \mathbb{R}$: $k \cdot u \in W$

First test, always. Take $k = 0$ in condition 3: $0 \cdot u = 0 \in W$. Every subspace contains the zero vector. If $0 \notin W$, stop: it is not a subspace. If $0 \in W$, you still have to check conditions 2 and 3.

Worked example. $W = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0\}$

- $(0, 0, 0)$ satisfies $0 + 0 + 0 = 0$, so $W \neq \varnothing$.
- If $u_1 + u_2 + u_3 = 0$ and $w_1 + w_2 + w_3 = 0$, then $(u_1 + w_1) + (u_2 + w_2) + (u_3 + w_3) = 0 + 0 = 0$. Closed under sum.
- $ku_1 + ku_2 + ku_3 = k{\cdot}0 = 0$. Closed under scalar product.

So $W \leq \mathbb{R}^3$, a plane through the origin. Change the equation to $x + y + z = 1$ and it fails at once: $(0, 0, 0)$ gives $0 \neq 1$.



Plane through 0
Subspace
x + y + z = 0

Line through 0
Subspace
all multiples k·u

Plane missing 0
Not a subspace
x + y + z = 1

Left and middle: the green result ($u + w$ or $k{\cdot}u$) stays inside. Right: u and w end on the plane, but $u + w$ escapes, because the plane never contained 0.

SUBSPACE TESTER · PICK A SET

$x+y+z=0$ | $x+y+z=1$ | $z=0$ | $y=2x$ | $x{\cdot}y=0$ | $x \geq 0$

$W = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0\}$

✓ **Contains 0? Yes**
(0, 0, 0) gives 0 + 0 + 0 = 0.

✓ **Closed under sum? Yes**
If $u_1 + u_2 + u_3 = 0$ and $w_1 + w_2 + w_3 = 0$, the components of u + w add up to 0 + 0 = 0.

✓ **Closed under scalar product? Yes**
$ku_1 + ku_2 + ku_3 = k{\cdot}0 = 0$.

Subspace A plane through 0.

Extra In $\mathbb{R}^3$ the only subspaces are $\{0\}$, lines through 0, planes through 0, and $\mathbb{R}^3$ itself.

STEP 3 · USES STEP 1

Linear combination

The only way vectors get built from other vectors.

Intuition. Vectors are ingredients and scalars are amounts. A linear combination is a recipe: take some of each, scaled, and add.

DEFINITION

$v \in V$ is a linear combination of $u_1, \ldots, u_n$ if there are real numbers $k_1, \ldots, k_n$ such that

$$v = k_1 \cdot u_1 + k_2 \cdot u_2 + \ldots + k_n \cdot u_n$$

Examples.

- $(1, -3) = 1 \cdot (1, 0) + (-3) \cdot (0, 1)$
- $(2, -3, 1) = 2 \cdot (1, 0, 0) + (-1) \cdot (0, 3, 1) + 2 \cdot (0, 0, 1)$. Check: $(2, 0, 0) + (0, -3, -1) + (0, 0, 2) = (2, -3, 1)$.

IN CODE

```
const combine = (us: Vec[], k: number[]): Vec =>
  us.reduce((acc, u, i) => add(acc, scale(k[i], u)), zero);
```

STEP 4 · USES STEPS 2 AND 3

Generated subspace and generating system

Everything you can cook from a fixed set of ingredients.

DEFINITION

The vector space generated by $\{u_1, \ldots, u_n\}$ is the set of all their linear combinations:

$$\langle u_1, \ldots, u_n \rangle = \{\, k_1 \cdot u_1 + \ldots + k_n \cdot u_n \mid k_i \in \mathbb{R} \,\}$$

It is always a subspace of V. The set $\{u_1, \ldots, u_n\}$ is called a generating system of it. A generating system of V means $\langle u_1, \ldots, u_n \rangle = V$: every vector of V is reachable.

Example. $\langle (1, -1, 3), (2, -5, 6) \rangle = \{(k + 2h, -k - 5h, 3k + 6h) \mid k, h \in \mathbb{R}\}$

How to test whether $v \in \langle u_1, \ldots, u_n \rangle$. Write $v = k_1 u_1 + \ldots + k_n u_n$ and solve the system for the k_i. A solution exists means yes.

Worked. Is $(3, 1, 4) \in \langle (1, 0, 1), (1, 1, 1) \rangle$? $k(1, 0, 1) + h(1, 1, 1) = (k + h, h, k + h)$. The first and third components need $k + h = 3$ and $k + h = 4$ at the same time. Impossible, so no.

Example. $\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 0)\}$ generates $\mathbb{R}^3$, because $(u_1, u_2, u_3) = u_1(1, 0, 0) + u_2(0, 1, 0) + u_3(0, 0, 1)$. But $(1, 1, 0)$ is redundant: the first three already reach everything. That redundancy is step 5.

STEP 5 · USES STEPS 3 AND 4

Linear dependence and independence

Is any vector in the set redundant?

Intuition. A set is linearly dependent if at least one vector is redundant: you can build it from the others, so removing it does not shrink what the set generates. Linearly independent means every vector adds a genuinely new direction. Spreadsheet analogy: a column defined as =A+B adds no new information.

DEFINITION

$u_1, \ldots, u_n$ are linearly dependent if some u_i is a linear combination of the others. Equivalently, if

$$k_1 \cdot u_1 + \ldots + k_n \cdot u_n = 0$$

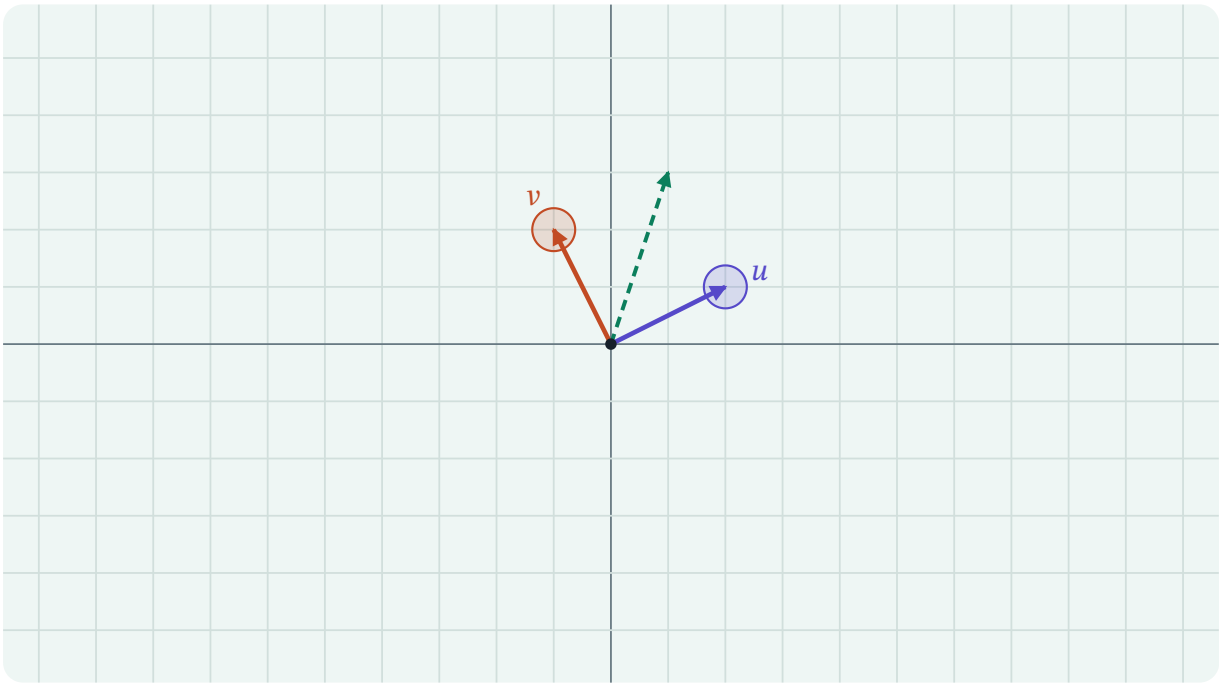
has a non-trivial solution (not all $k_i = 0$). They are linearly independent if the only solution is $k_1 = \ldots = k_n = 0$.

Examples.

- $(3, 3, 2), (1, 1, -1), (2, 2, 3)$: $(3, 3, 2) = 1 \cdot (1, 1, -1) + 1 \cdot (2, 2, 3)$, so $1 \cdot (3, 3, 2) - 1 \cdot (1, 1, -1) - 1 \cdot (2, 2, 3) = 0$. Non-trivial, so dependent.
- $(1, 0), (0, 1)$: $k(1, 0) + h(0, 1) = (k, h) = (0, 0)$ forces $k = h = 0$. Independent.

Extra Geometric reading: in $\mathbb{R}^2$, two vectors are dependent exactly when they are parallel. In $\mathbb{R}^3$, three vectors are dependent exactly when they lie in one plane through 0.

PLAYGROUND · $\mathbb{R}^2$ · STEPS 3 TO 6 AT ONCE



Drag the circles; they snap to the grid. Dashed green arrow: $k \cdot u + h \cdot v$. Green shading: what the set generates.

k 1

h 1

Independent | Parallel | Contains 0 | Add a third vector w

Linearly independent **Rank 2 of 2 vectors**

`u = (2, 1) v = (-1, 2) | k·u + h·v = 1·(2, 1) + 1·(-1, 2) = (1, 3)`

⟨u, v⟩ = $\mathbb{R}^2$. The set is a generating system of $\mathbb{R}^2$: every point is some combination.

Only k = h = 0 gives k·u + h·v = 0, so neither vector is redundant.

STEP 6 · USES STEP 5

Rank of a set of vectors

How many genuinely different directions the set contains.

DEFINITION

The rank of a set of vectors is the maximum number of linearly independent vectors in it.

Intuition. Throw away redundant vectors until none is left. What remains is the rank. It equals the dimension of what the set generates: $\dim \langle u_1, \ldots, u_n \rangle = \text{rank}$.

Examples.

- $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ has rank 3.
- $\{(1, 2, 3), (0, 2, 2), (1, 4, 5)\}$ has rank 2, because $(1, 4, 5) = (1, 2, 3) + (0, 2, 2)$.
- $\{(1, 0, 2), (2, 0, 4), (3, 0, 6)\}$ has rank 1: all three are multiples of the first.

Link to step 5. A set of n vectors is independent $\Leftrightarrow$ its rank is n. It is dependent $\Leftrightarrow$ its rank is less than n.

Preview of next week. Put the vectors as rows or columns of a matrix A. The rank of the set is $\text{rk}(A)$, the order of its largest non-zero minor. For n vectors in $\mathbb{R}^n$: independent $\Leftrightarrow$ $\det \neq 0$.

RANK LAB · VECTORS IN $\mathbb{R}^3$

Canonical basis | One is a sum | All multiples | Hidden combination | Four vectors

$u_1 = ($ 1 2 3 $)$

$u_1 = ($ 1 , 2 , 3 $)$

$u_2 = ($ 0 , 2 , 2 $)$

$u_3 = ($ 1 , 4 , 5 $)$

Add a vector | Remove the last one

Rank 2 Linearly dependent

- $\langle u_1, u_2, u_3 \rangle$ is a plane through 0 (dimension 2).
- Generating system of $\mathbb{R}^3$? No: that needs rank 3.
- Basis of $\mathbb{R}^3$? No: the vectors are dependent.
- A basis of what they generate: $\{u_1, u_2\}$

REDUNDANT VECTORS, SCANNING LEFT TO RIGHT

- $u_3 = 1 \cdot u_1 + 1 \cdot u_2$

STEP 7 · USES STEPS 4 AND 5

Basis and dimension

A generating system with nothing redundant.

DEFINITION

$B = \{u_1, \ldots, u_n\}$ is a basis of V if

1. it is a generating system of V, and
2. its vectors are linearly independent.

Then the dimension of V is n, written $\dim(V) = n$. This makes sense because all bases of V have the same number of vectors. The same definition works for a subspace W.

Canonical basis. In $\mathbb{R}^n$: $e_1 = (1, 0, \ldots, 0)$, …, $e_n = (0, \ldots, 0, 1)$. Not the only one: $\{(1, 1, 0), (0, 2, 0), (1, 0, 4)\}$ is also a basis of $\mathbb{R}^3$.

PROPOSITION · IF DIM(V) = N

1. Every generating system of V has at least n vectors.
2. Every linearly independent set of V has at most n vectors.
3. n linearly independent vectors form a basis.
4. A generating system with n vectors is a basis.

So two vectors can never generate $\mathbb{R}^3$, and four vectors in $\mathbb{R}^3$ are always dependent.

Components in a basis. Each $v \in V$ is written in exactly one way as $v = c_1u_1 + \ldots + c_nu_n$. The numbers $c_1, \ldots, c_n$ are the components of v in B.

Example. In $B = \{(1, 0), (2, 1)\}$: $(-2, 3) = c_1(1, 0) + c_2(2, 1) = (c_1 + 2c_2, c_2)$, so $c_2 = 3$ and $c_1 = -8$. In the canonical basis the components are just -2 and 3.

Design analogy. In RGB, every color is $r{\cdot}(1, 0, 0) + g{\cdot}(0, 1, 0) + b{\cdot}(0, 0, 1)$: red, green and blue are the canonical basis of $\mathbb{R}^3$, and (r, g, b) are the components. A palette of red, green, blue and purple still generates every color, but it is not a basis, because purple = red + blue is redundant. Analogy (it ignores the 0 to 255 clamp).

REVIEW

What you need to know

Tick items off as you master them. Your ticks stay in this browser.

0 of 22 checked Clear ticks

Definitions to state

- [ ] Real vector space $(V, +, \cdot)$
- [ ] Vector subspace $W \leq V$ (3 conditions)
- [ ] Linear combination
- [ ] Generated subspace $\langle u_1, \ldots, u_n \rangle$
- [ ] Generating system
- [ ] Linear dependence and independence
- [ ] Rank of a set of vectors

Skills the exercises test

- [ ] Decide whether a set is a subspace (zero test first, then closure)
- [ ] Decide whether $v \in \langle u_1, \ldots, u_n \rangle$ by solving for the coefficients
- [ ] Decide dependence or independence (homogeneous system now, determinant next week)
- [ ] Find the rank of a set and the dimension it generates

- [ ] Basis and dimension
- [ ] Canonical basis of $\mathbb{R}^n$
- [ ] Components of a vector in a basis
- [ ] Decide whether a set is a basis of $\mathbb{R}^n$
- [ ] Compute components of a vector in a basis

Facts to memorize

- [ ] Every subspace contains 0
- [ ] Generating system of V: at least $\dim V$ vectors
- [ ] Independent set of V: at most $\dim V$ vectors
- [ ] n independent vectors in a space of dimension n form a basis
- [ ] A generating system with n vectors is a basis
- [ ] Rank of a set = dimension of what it generates

Many exercises on these ideas are solved with determinants and the rank of a matrix, which come next week.

PRACTICE

Check questions

Answer on paper first, then open the solution.

Q1 Is $W = \{(x, y) \in \mathbb{R}^2 \mid y = 2x + 1\}$ a subspace of $\mathbb{R}^2$? Justify in one line. Step 2

Show answer